# AKSHIT SHARMA

AI/ML Engineer · LLM Infrastructure · Agentic Systems
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## SUMMARY

B.Tech CSE (Data Science, AI & ML), Bahra University, Class of 2026. I build GenAI systems: Retrieval-Augmented Generation (RAG) pipelines, LangGraph multi-agent setups, and FastAPI/Docker deployments on GCP and AWS. Headless_BAI held up when validated across 1,000+ real user sessions. GeoSmart runs under 420MB RAM at a 15–40x speedup over plain Python.

## TECHNICAL SKILLS

**Languages:** Python, C++, Julia, TypeScript
**GenAI/LLM & Retrieval:** RAG pipelines (ingestion, chunking, embeddings, hybrid search), prompt engineering, fine-tuning (QLoRA), LangGraph (multi-agent orchestration), LangChain, hallucination mitigation & output validation
**AI Testing & Evaluation:** Model/system evaluation pipelines, K-Means-based behavioral cohort validation (WCSS), latency and accuracy benchmarking, telemetry-driven regression testing
**MLOps & Cloud:** Docker, GCP Cloud Run, AWS, IBM Cloud, CI/CD, Git, hybrid cloud deployment
**Software Engineering:** React/TypeScript, FastAPI, Flask, REST API design, OOP design, n8n automation
**ML/AI:** TensorFlow/Keras, PyTorch, scikit-learn, LSTM, NLP (LLM and non-LLM approaches), K-Means clustering, LLM routing & inference optimization, model serving
**Data & Vector Stores:** Qdrant, Supabase, MySQL, pandas, NumPy

## EXPERIENCE

**AI/ML Engineering Trainee** — *Future Finders* *Jan – Jun 2026*
- Built and validated deep learning models end-to-end — data preprocessing, training, evaluation, and deployment — following engineering best practices for code quality and testing
- Engineered reusable model training pipelines and validation frameworks using Python (OOP design), standardizing experimentation and evaluation across projects
- Collaborated using Git-based version control and code review workflows

**Deep Learning Intensive** — *PDPM IIITDM Jabalpur* *Jun 2025*
- Completed an intensive deep learning curriculum covering advanced neural network architectures, training methodologies, and real-world deployment strategies; ranked top 5% of 100+ participants (Excellence Award)

**Cloud ML Deployment** — *IBM Cloud Trainee* *Sep – Oct 2024*
- Deployed containerized ML applications with auto-scaling policies on IBM Cloud, gaining hands-on production infrastructure exposure

## KEY PROJECTS

**Headless_BAI – AI Layout Optimization Engine** *Jan – Jun 2026*
- Built a custom-compiled Chromium C++ binary instrumented with behavioral telemetry, routing UI decisions through Llama-3-8B (via Groq) for deterministic, prompt-driven JSON layout generation (<500ms inference, T=0.1)
- Designed an evaluation framework validating 3 behavioral cohorts via K-Means clustering (WCSS: 23.24) against real-time telemetry
- Built an end-to-end telemetry pipeline (sensor.js → FastAPI → Supabase) processing 10,800+ behavioral events in real time
- Achieved a 31.42% session-weighted reduction in UI spatial friction, validated across 1,000+ user sessions

**DeepResearch AI Agent – Multi-Agent RAG System** *Feb – Mar 2026*
- Designed and built an agentic RAG system for knowledge retrieval using LangGraph for multi-agent orchestration (research, analysis, and memory agents)
- Integrated Qdrant for high-performance semantic search against document embeddings, mitigating hallucinations and improving context retrieval accuracy
- Deployed via Python/FastAPI, Docker, and CI/CD on GCP Cloud Run, achieving <900ms end-to-end inference latency

**GeoSmart Advisor – Geospatial Site Selection Intelligence** *Jun – Sep 2025*
- Engineered a cloud-agnostic intelligence system using a locally-hosted, 4-bit quantized Phi-3-mini SLM for narrative report generation (<900ms latency)
- Developed a Julia computational core (1.10) for spatial matrix operations, achieving a 15–40x speedup over Python (<180ms) while capping peak RAM at 420MB
- Implemented a React/deck.gl (WebGL) frontend with a custom asymmetric loss function (1.5x penalty for overconfidence) for conservative, defensible site recommendations

## HONORS

- **Excellence Award, Deep Learning Summer School** — PDPM IIITDM Jabalpur (top 5% of 100+ participants) | Jun 2025
- **Google Student Ambassador** — Ran sessions for college students on Google Gemini and applied AI use cases | Dec 2025

## EDUCATION

B.Tech Computer Science Engineering (Data Science, AI & ML) — Bahra University | Class of 2026

## CERTIFICATIONS

Geo Processing with Python – ISRO (Jun 2023) | Full Stack Web Development – Amazon/Coursera (Feb 2025) | AWS Educate ML Foundations (Jun 2025) | Cyber Security – IBM SkillsNetwork (Mar 2026)